

Project Title: Smart Study Buddy

Objective: Build a beginner-friendly GenAI application using Google Gemini, LangChain, and Streamlit. The application generates complete study material from a topic entered by the user.

Learning Outcomes

- Understand Google Gemini API integration.
- Learn LangChain basics.
- Build a Streamlit application.
- Practice Prompt Engineering.
- Generate structured AI responses.

Project Features

- Topic input
- Difficulty selection (Easy/Medium/Hard)
- Explanation
- Simple example
- Real-life example
- 5 MCQs
- 5 Interview Questions
- Summary

Technology Stack

- Python 3.11+
- Google Gemini API
- LangChain
- Streamlit
- python-dotenv

Folder Structure

- Smart_Study_Buddy/
 - |— app.py
 - |— prompts.py
 - |— .env
 - |— requirements.txt
 - |— README.md

Application Workflow

- 1. User enters topic.
- 2. User selects difficulty.
- 3. Streamlit sends prompt to Gemini through LangChain.
- 4. Gemini generates structured study material.
- 5. Output is displayed in sections.

Prompt Template

- You are an expert teacher.
- Teach the given topic according to the selected difficulty.
- Generate:
 - 1. Explanation
 - 2. Simple Example
 - 3. Real-life Example
 - 4. Five MCQs with answers
 - 5. Five Interview Questions
 - 6. Summary

Student Tasks

- Create Streamlit UI
- Connect Gemini API
- Store API key in .env
- Create prompt template
- Display structured output
- Test with different topics

Sample Input

- Topic: Python Functions
- Difficulty: Easy

Sample Output

- Explanation
- Example
- Real-life Example
- 5 MCQs
- 5 Interview Questions
- Summary

Evaluation (10 marks)

- UI - 2
- Gemini Integration – 2
- Prompt Engineering – 2
- Structured Output - 1

- Code Quality - 2
- Documentation - 1

Future Enhancements

- PDF Export
- Voice Input
- Quiz Download
- Chat History
- Dark Mode

Smart Study Buddy
GenAI

BRING ON YOUR CODING ATTITUDE